Answers and talk notes

Write and Evaluate Numerical Expressions with Exponents

What your student is learning

Tonight we evaluate expressions that mix grouping symbols, powers and arithmetic — in the one order that gives one answer.

Answers

Ask your student to explain before you look.

1. **D.** Add 2 + 4. Add 2 + 4 first. Grouping symbols are done before multiplying.
2. **B.** 10. $(8 - 3) \times 2 = 5 \times 2 = 10$.
3. **A.** 64. $(6 + 2)^2 = 8^2 = 64$. Add inside first, then square.
4. **C.** 4. $20 - (3 + 1)^2 = 20 - 4^2 = 20 - 16 = 4$.
5. **B.** 14. $2 + 3 \times (5 - 1) = 2 + 3 \times 4 = 2 + 12 = 14$.
6. **D.** 24. $6 + 2 \times (7 - 4)^2 = 6 + 2 \times 3^2 = 6 + 2 \times 9 = 6 + 18 = 24$.

Talk about it

What did you do first, second and third, and why does that order give only one answer?

Home mission

Make a game score at the table. Start with 6 points, then add 2 rounds worth $(6 - 3)^2$ points each. Ask your student to say the expression and rewrite it after every step.

When they are stuck, ask

- What step comes first, and why?
- Can you rewrite the whole expression after that step?
- These two operations have the same rank — which way do we go?

Try not to say

- Just work left to right, it usually comes out.
- Multiplication always comes before division.
- Do the exponent last since it is a small number.